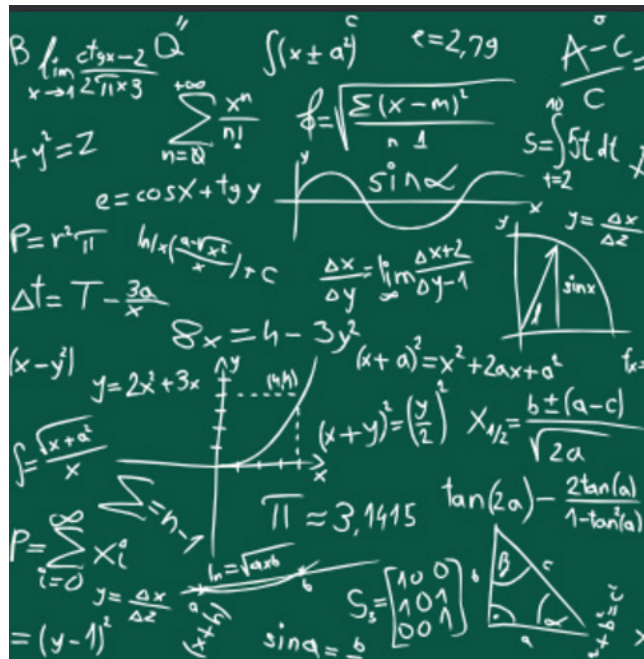


Rules for computing derivatives



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Rules for computing derivatives

- We now look at the relation between the derivative operation and the main operations on functions that we already know; in particular we will show the relation between:
 1. differentiation and algebraic operations
 2. differentiation and composition
 3. differentiation and inversion

1.1 Algebra of derivatives

Theorem 1: algebra of derivatives

Let $f, g: (a, b) \rightarrow \mathbb{R}$ be two functions differentiable in (a, b) .

Then $f \pm g, f \cdot g, f/g$ ($g \neq 0$) are differentiable in (a, b) and the following formulas hold:

$$(f \pm g)' = f' \pm g' \quad (1)$$

$$(f \cdot g)' = f' \cdot g + f \cdot g' \quad (2)$$

$$\left(\frac{f}{g}\right)' = \frac{f' \cdot g - f \cdot g'}{g^2} \quad (g \neq 0) \quad (3)$$

- From rule (2) we deduce

$$(k \cdot g)' = k \cdot g' \quad \text{with } k \in \mathbb{R} \text{ constant} \quad (4)$$

since the derivative of a constant is equal to zero.

- From rule (3) we deduce

$$\left(\frac{1}{g}\right)' = -\frac{g'}{g^2} \quad (g \neq 0) \quad (5)$$

- Rule (2) is called the **Leibniz rule** (product rule) and it extends to the product of n functions:

$$(f_1 f_2 \cdots f_n)' = f_1' f_2 \cdots f_n + f_1 f_2' \cdots f_n + \cdots + f_1 f_2 \cdots f_n' \quad (6)$$

- The theorem actually has a pointwise character: that is, if f and g are differentiable at a point $x_0 \in (a, b)$, then at that point $f \pm g$, $f \cdot g$, f/g are also differentiable, and the formulas above hold (and they extend to intervals).

Proof. We prove rule (2). Fixing $x \in (a, b)$, we have

$$\begin{aligned} & f(x+h)g(x+h) - f(x)g(x) = \\ &= f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x), \end{aligned}$$

and hence the difference quotient can be written as:

$$\begin{aligned} & \frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \\ &= f(x+h) \underbrace{\frac{g(x+h) - g(x)}{h}}_{=g'(x) \text{ for } h \rightarrow 0} + g(x) \underbrace{\frac{f(x+h) - f(x)}{h}}_{=f'(x) \text{ for } h \rightarrow 0} \end{aligned}$$

We have

$$f(x+h) \rightarrow f(x) \text{ for } h \rightarrow 0,$$

since f is continuous, being differentiable.

Hence

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} \rightarrow f(x)g'(x) + f'(x)g(x) \text{ for } h \rightarrow 0.$$

□

Proof. We prove rule (5). Fixing $x \in (a, b)$, the difference quotient can be written as:

$$\begin{aligned} & \frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] = \frac{g(x) - g(x+h)}{h g(x) g(x+h)} = \\ &= - \frac{g(x+h) - g(x)}{h} \cdot \frac{1}{g(x) g(x+h)} \end{aligned}$$

We have

$$g(x+h) \rightarrow g(x) \text{ for } h \rightarrow 0,$$

since g is continuous, being differentiable.

Hence

$$\frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] \rightarrow - \frac{g'(x)}{g^2(x)} \text{ for } h \rightarrow 0.$$

□

Proof. Note that rule (3) follows from rules (2) and (5); indeed:

$$\begin{aligned} \left(\frac{f(x)}{g(x)}\right)' &= \left(f(x) \cdot \frac{1}{g(x)}\right)' = \underbrace{f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{g(x)}\right)'}_{\text{rule (2)}} = \\ &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \underbrace{\left(-\frac{g'(x)}{g^2(x)}\right)}_{\text{rule (5)}} = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g^2(x)}. \end{aligned}$$

□

Remark 1

Given the function $f(x) = \tan x$, the derivative function is $f'(x) = \frac{1}{\cos^2 x} = 1 + \tan^2 x$.

Proof. Using formula (3) we have

$$f'(x) = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = 1 + \tan^2 x$$

□

Remark 2

Given the function $f(x) = \cot x$, the derivative function is $f'(x) = -\frac{1}{\sin^2 x} = -(1 + \cot^2 x)$.

Proof. Using formula (3) we have

$$\begin{aligned} f'(x) &= \left(\frac{\cos x}{\sin x}\right)' = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\ &= -\frac{1}{(\sin^2 x)} = -(1 + \cot^2 x) \end{aligned}$$

□

1.2 Derivative of a composite function

Theorem 2: chain rule

Let $g \circ f$ be the composite function of two functions f and g . If f is differentiable at a point x and g is differentiable at $y = f(x)$, then $g \circ f$ is differentiable at x and the following formula holds:

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) \quad (7)$$

Proof. We have

$$(g \circ f)(x+h) - (g \circ f)(x) = g(f(x+h)) - g(f(x))$$

If we set

$$k = f(x+h) - f(x) \quad \text{and} \quad y = f(x) \quad \text{then} \quad f(x+h) = y + k$$

By the continuity of f , $h \rightarrow 0$ implies $k \rightarrow 0$. With the new notation we have:

$$g(f(x+h)) - g(f(x)) = g(y+k) - g(y)$$

We now observe that the definition of derivative:

$$g'(y) = \lim_{k \rightarrow 0} \frac{g(y+k) - g(y)}{k}$$

can be rewritten, for $k \neq 0$, as

$$\frac{g(y+k) - g(y)}{k} = g'(y) + \varepsilon(k)$$

where $\varepsilon(k)$ denotes a quantity that tends to zero as $k \rightarrow 0$.

Multiplying both sides of the previous equation by k we find

$$g(y+k) - g(y) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

a relation that also holds for $k = 0$. Therefore:

$$g(f(x+h)) - g(f(x)) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

Dividing by h , and observing that

$$\frac{k}{h} \rightarrow f'(x) \quad \text{for} \quad h \rightarrow 0$$

which implies, as seen above, also $k \rightarrow 0$, we obtain the claim. □

- As the name “chain rule” suggests, (7) can be generalized to the composition of any number of functions, composed one with another. For example, for three functions we have:

$$\left(f(g(h(x))) \right)' = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Example 1: Derivative of a composite function

We compute the derivative function of the composite function:

$$h(x) = \underbrace{\sin^3 x}_{=(g \circ f)(x)}$$

The two functions are:

$$f(x) = \sin x \quad \text{and} \quad g(y) = y^3 \quad \text{with} \quad y = \sin x.$$

Both functions are differentiable on all of $\mathbb{R}$. We have

$$g'(y) = 3y^2 \quad \text{and} \quad f'(x) = \cos x$$

hence, using the chain rule (7) and substituting back, we obtain:

$$h'(x) = 3 \sin^2 x \cdot \cos x.$$

Consider the composite function $g(f(x))$, equivalently written $(g \circ f)(x)$. Setting $y = f(x)$ and $w = g(y)$, and using the (Leibniz) notation

$$\frac{df}{dx} \quad \text{and} \quad \frac{dg}{dy} \quad \text{for the derivatives of } f \text{ and } g \text{ with respect to } x$$

the (chain) rule (7) can be rewritten as:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} \tag{8}$$

Rule (8) expresses the fact that the rate of change of w with respect to x is the product of the “intermediate” rates of change, of w with respect to y and of y with respect to x .

Example 2: Derivative of a composite function

Consider the composite function of the previous exercise. Setting

$$y = \sin x \quad (y = f(x)) \quad \text{and} \quad w = y^3 \quad (w = g(y))$$

and using Leibniz's notation, rule (8) would be written:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} = 3y^2 \cdot \cos x.$$

And hence, substituting $y = \sin x$, we obtain:

$$\frac{dw}{dx} = 3 \sin^2 x \cdot \cos x.$$

Example 3: Derivative of a composite function

We compute the derivative function of the composite function (multiplied by a constant):

$$h(x) = A \cdot \underbrace{\sin(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with} \quad A, \omega, \varphi \in \mathbb{R}.$$

The two functions are:

$$f(x) = \omega x + \varphi \quad \text{and} \quad g(y) = \sin y \quad \text{with} \quad y = \omega x + \varphi.$$

Both functions are differentiable on all of $\mathbb{R}$. We have:

$$f'(x) = \omega \quad \text{and} \quad g'(y) = \cos y$$

hence, using rule (7), rule (4) and substituting back, we obtain:

$$h'(x) = A \cdot \left(\cos(\omega x + \varphi) \cdot \omega \right)$$

Example 4: Derivative of a product of composite functions

We compute the derivative function of the product of two composite functions times a constant:

$$h(x) = A \cdot \underbrace{e^{-\alpha x}}_{=(r \circ s)(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with } A, \omega, \varphi \in \mathbb{R}, \alpha \in \mathbb{R}_+.$$

The functions are:

$$s(x) = -\alpha x \quad \text{and} \quad r(z) = e^z \quad \text{with } z = -\alpha x; \quad f(x) = \omega x + \varphi \quad \text{and} \quad g(y) = \cos y \quad \text{with } y = \omega x + \varphi.$$

The functions are differentiable on all of $\mathbb{R}$. We have:

$$s'(x) = -\alpha \quad \text{and} \quad r'(z) = e^z; \quad f'(x) = \omega \quad \text{and} \quad g'(y) = -\sin y$$

consequently, using the chain rule (7) and substituting back, we obtain:

$$(r \circ s)'(x) = -\alpha e^{-\alpha x}; \quad (g \circ f)'(x) = -\sin(\omega x + \varphi) \cdot \omega$$

Hence, using rule (2) and rule (4):

$$h'(x) = A \cdot \left(\underbrace{-\alpha e^{-\alpha x}}_{(r \circ s)'(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{(g \circ f)(x)} + \underbrace{e^{-\alpha x}}_{(r \circ s)(x)} \cdot \underbrace{(-\sin(\omega x + \varphi) \cdot \omega)}_{(g \circ f)'(x)} \right)$$

Example 5: Derivative of a composite function

Let $f(x) > 0$ be differentiable; we compute the derivative of the function

$$h(x) = \log(f(x))$$

using the chain rule (7) we have:

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

Remark 3

Given the function $f(x) = a^x$, with $a > 0$, the derivative function is $f'(x) = a^x \cdot \log a$.

Proof. We have

$$f'(x) = \left(\exp(x \cdot \log a) \right)'$$

hence, using the formulas for the derivative of e^y (with $y = x \log a$) and for the derivative of a composite function, we have

$$f'(x) = \exp(x \cdot \log a) \cdot \log a = a^x \cdot \log a$$

□

Remark 4

Given the function $f(x) = \log_a x$, with $a > 0$, $a \neq 1$, the derivative function is $f'(x) = \frac{1}{x \log a}$.

Proof. We have

$$f'(x) = \left(\frac{\log x}{\log a} \right)'$$

hence, using the formulas for the derivative of $\log x$ and for the derivative of $k f(x)$, we have:

$$f'(x) = \frac{1}{\log a} \cdot \frac{1}{x} = \frac{1}{x \log a}$$

□

Remark 5

Given the function $f(x) = \sinh x$, the derivative function is $f'(x) = \cosh x$.

Proof. We have:

$$f'(x) = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{1}{2} \left(e^x - (-e^{-x}) \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

□

Remark 6

Given the function $f(x) = \cosh x$, the derivative function is $f'(x) = \sinh x$.

Proof. We have:

$$f'(x) = \left(\frac{e^x + e^{-x}}{2} \right)' = \frac{1}{2} \left(e^x + (-e^{-x}) \right) = \frac{e^x - e^{-x}}{2} = \sinh x$$

□

The derivatives of functions of the type:

$$h(x) = f(x)^{g(x)}$$

are based on rewriting the function (with $f(x) > 0$) in the following form:

$$f(x)^{g(x)} = \exp \left(g(x) \cdot \log(f(x)) \right)$$

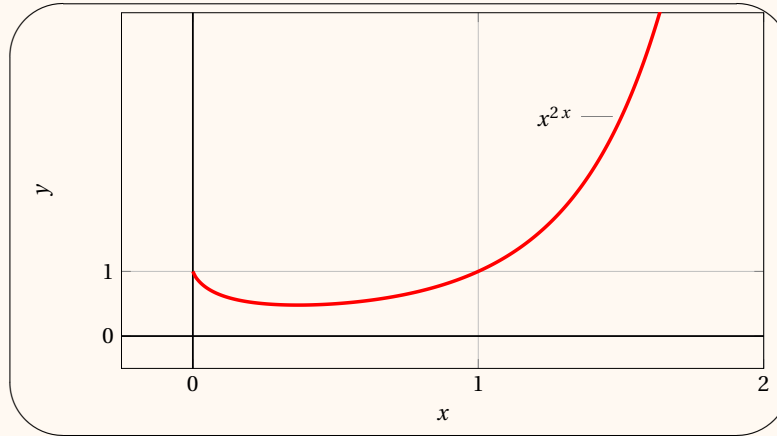
We have:

$$\begin{aligned} \left(f(x)^{g(x)} \right)' &= \left(\exp \left(g(x) \cdot \log(f(x)) \right) \right)' \\ &= \exp \left(g(x) \cdot \log(f(x)) \right) \cdot \left(g(x) \cdot \log(f(x)) \right)' \\ &= f(x)^{g(x)} \cdot \left(g'(x) \cdot \log(f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \right) \end{aligned}$$

Example 6: Derivative of a function raised to another function

We compute the derivative of the function

$$h(x) = x^{2x} = \exp(2x \cdot \log x) \text{ with } x > 0$$



The two functions are:

$$f(x) = x \quad \text{and} \quad g(x) = 2x$$

Hence:

$$h'(x) = \left(\exp(2x \cdot \log x) \right)' = \exp(2x \cdot \log x) \cdot (2x \cdot \log x)' = x^{2x} \cdot \left(2 \log x + 2x \frac{1}{x} \right) = x^{2x} (2 \log x + 2)$$

Consider the absolute value of a function: $|f(x)|$. At the points where $f(x) \neq 0$, the differentiation of a composite function gives:

$$(|f(x)|)' = \operatorname{sgn}(f(x)) \cdot f'(x) = \begin{cases} f'(x) & \text{for } f(x) > 0 \\ -f'(x) & \text{for } f(x) < 0 \end{cases}$$

In general, we expect the function $|f(x)|$ to have corner points at the points where $f(x)$ vanishes.

Example 7: Derivative of a function with absolute value

We compute the derivative function of the function:

$$f(x) = |x^2 - 4x + 3|, \text{ with } x \neq 1 \text{ and } x \neq 3, \text{ we have } f'(x) = \operatorname{sgn}(x^2 - 4x + 3) \cdot (2x - 4)$$

equivalently:

$$f'(x) = \begin{cases} 2x - 4, & x < 1 \text{ or } x > 3 \quad \text{i.e. for } f(x) > 0 \\ -(2x - 4) & 1 < x < 3 \quad \text{i.e. for } f(x) < 0 \end{cases}$$

At $x_0 = 1$, we need to use the limit of the difference quotient:

$$\frac{f(x_0 + h) - f(x_0)}{h} = \frac{f(1 + h) - f(1)}{h} = \frac{|(1 + h)^2 - 4(1 + h) + 3|}{h} = \frac{|h^2 - 2h|}{h}$$

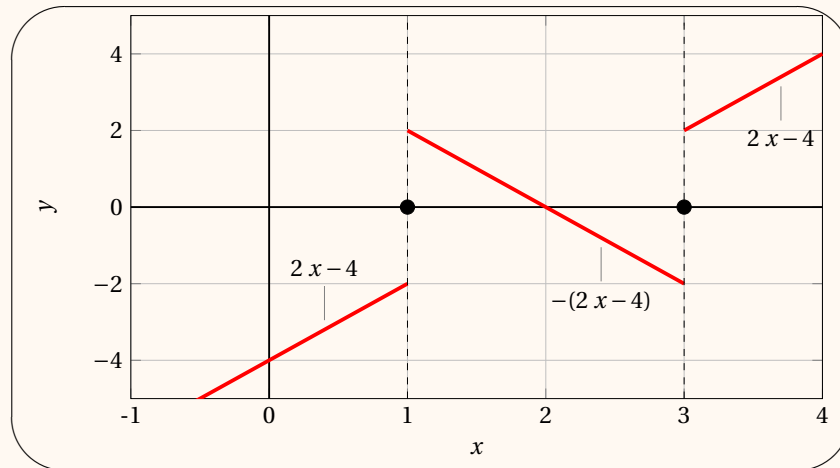
If $h \rightarrow 0^-$ we have $|h^2 - 2h| = h^2 - 2h$, hence:

$$\lim_{h \rightarrow 0^-} \frac{h^2 - 2h}{h} = -2$$

If $h \rightarrow 0^+$ we have $|h^2 - 2h| = -h^2 + 2h$, hence:

$$\lim_{h \rightarrow 0^+} \frac{-h^2 + 2h}{h} = 2$$

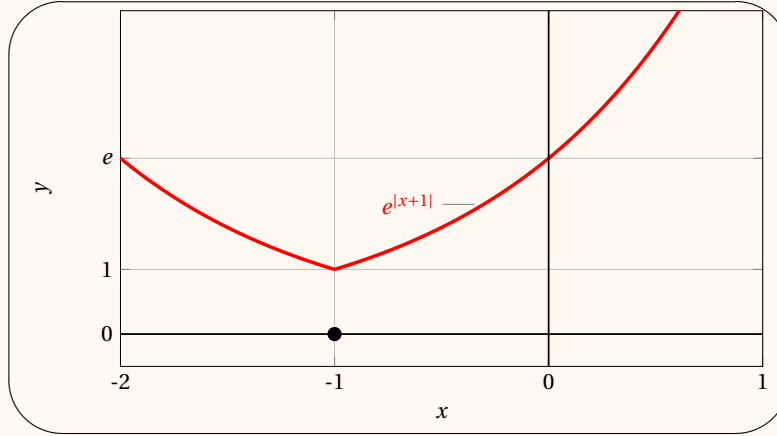
The function is not differentiable at $x_0 = 1$ and $x_0 = 3$ (by arguments analogous to those for $x_0 = 1$), where it has corner points.



Example 8: Derivative of a function with absolute value

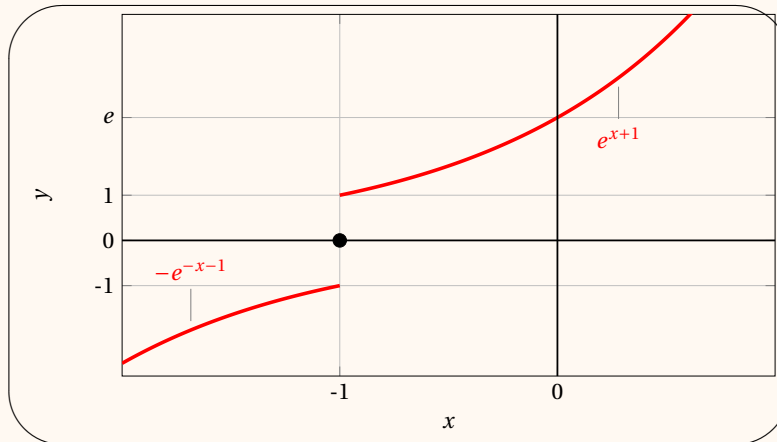
We compute the derivative of the function:

$$f(x) = e^{|x+1|}$$



For $x \neq -1$, we have:

$$f(x) = \begin{cases} e^{x+1}, & x > -1 \\ e^{-x-1}, & x < -1 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} e^{x+1}, & x > -1 \\ -e^{-x-1}, & x < -1 \end{cases}$$



At $x_0 = -1$, we need to use the limit of the difference quotient:

$$\frac{f(x_0 + h) - f(x_0)}{h} = \frac{f(-1 + h) - f(-1)}{h} = \frac{e^{|-1+h+1|} - 1}{h} = \frac{e^{|h|} - 1}{h}$$

We have:

$$\text{if } h \rightarrow 0^+, e^{|h|} = e^h \text{ and } \lim_{h \rightarrow 0^+} \frac{e^h - 1}{h} = 1; \quad \text{if } h \rightarrow 0^-, e^{|h|} = e^{-h} \text{ and } \lim_{h \rightarrow 0^-} \frac{e^{-h} - 1}{h} = -1$$

We conclude that, since the limit of the difference quotient does not exist, the function is not differentiable at $x_0 = -1$. The function has a corner point at $x_0 = -1$.

We now consider the derivatives of some logarithmic functions. We compute:

$$\left(\log(|x|) \right)' = \frac{1}{|x|} \cdot \operatorname{sgn}(x) = \frac{1}{x}$$

Moreover, by the differentiation of composite functions we have:

$$\left(\log(|f(x)|) \right)' = \frac{1}{|f(x)|} \cdot \operatorname{sgn}(f(x)) \cdot f'(x) = \frac{f'(x)}{f(x)}$$

Sometimes it is convenient to use the properties of logarithms to transform a logarithmic function before computing its derivative function:

$$(\log(cx))' = (\log(c) + \log(x))' = \frac{1}{x} \quad \text{with } c \in \mathbb{R}$$

$$\left(\log\left(\frac{ax+b}{cx+d}\right) \right)' = (\log(ax+b) - \log(cx+d))' = \frac{a}{ax+b} - \frac{c}{cx+d} \quad \text{with } a, b, c, d \in \mathbb{R}$$

1.3 Derivative of the inverse function

Theorem 3: derivative of the inverse function

Let $f : (a, b) \rightarrow \mathbb{R}$ be a continuous and invertible function in (a, b) and $g = f^{-1}$ its inverse function, defined in $f((a, b))$.

Assume moreover that $f'(x_0) \neq 0$ exists for some $x_0 \in (a, b)$.

Then g is differentiable at $y_0 = f(x_0)$ and

$$g'(y_0) = \frac{1}{f'(x_0)} \tag{9}$$

Proof. Let:

$$f(x_0) = y_0; \quad g(y_0) = x_0$$

$$f(x_0 + h) = y_0 + k; \quad g(y_0 + k) = x_0 + h$$

Consider the difference quotient of g at y_0 :

$$\frac{g(y_0 + k) - g(y_0)}{k} = \frac{h}{f(x_0 + h) - f(x_0)}$$

If $k \neq 0$, then $f(x_0 + h) - f(x_0) \neq 0$ and hence also $h \neq 0$; therefore the last quotient can also be rewritten in the form:

$$\frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}}$$

Moreover, for $k \rightarrow 0$ we have $g(y_0 + k) \rightarrow g(y_0)$ because g is continuous, being the inverse of a continuous function on an interval (theorem on the continuity of inverse functions); on the other hand $h = g(y_0 + k) - g(y_0)$, hence for $k \rightarrow 0$ also $h \rightarrow 0$, and by hypothesis

$$\frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}} \rightarrow \frac{1}{f'(x_0)} \quad \text{for } h \rightarrow 0$$

hence the following limit exists

$$\lim_{k \rightarrow 0} \frac{g(y_0 + k) - g(y_0)}{k} = \frac{1}{f'(x_0)}$$

□

- We observe that, assuming the differentiability of f^{-1} , (9) follows immediately from the identity

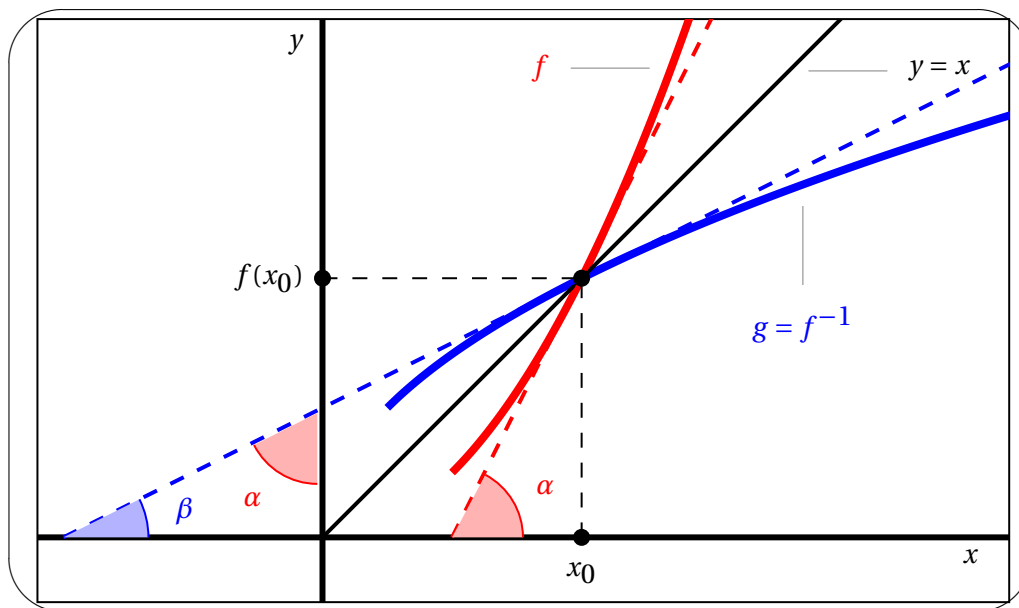
$$g(f(x)) = x$$

and from the chain rule:

$$g'(f(x)) \cdot f'(x) = 1$$

from which, if $f'(x) \neq 0$, (9) follows. The proof we have given, however, is necessary in order to deduce the differentiability of g from our hypotheses on f .

- (9) has a simple geometric meaning, recalling that the graphs of f and $g = f^{-1}$ are symmetric with respect to the bisector $y = x$. Geometrically we have:



hence the angles α and β are complementary, that is:

$$\left(\alpha + \beta = \frac{\pi}{2}\right)$$

and hence

$$f'(x) = \tan \alpha = \tan \left(\frac{\pi}{2} - \beta\right) = \frac{1}{\tan \beta} = \frac{1}{g'(y)}$$

With Leibniz's notation, setting $y = f(x)$ and $x = g(y)$, (9) is written in the form

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

- Pay attention to the fact that in the formula for the derivative of the inverse function, the derivatives f' and g' are computed at two different points: this is the main thing to be careful about when applying this theorem.

Remark 7

Given the function $f(y) = \arctan y$, the derivative function is $f'(y) = \frac{1}{1+y^2}$.

Proof. We set $y = \tan x$, $x = \arctan y$ with $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $y \in \mathbb{R}$. We have:

$$(\arctan y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + y^2}$$

□

Remark 8

Given the function $f(y) = \arcsin y$, the derivative function is $f'(y) = \frac{1}{\sqrt{1-y^2}}$

Proof. We set $y = \sin x$, $x = \arcsin y$, with $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $y \in [-1, 1]$. Since for those values of x we have

$$\cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - y^2}$$

we have:

$$(\arcsin y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}}$$

□

Remark 9

Given the function $f(y) = \arccos y$, the derivative function is $f'(y) = -\frac{1}{\sqrt{1-y^2}}$

Proof. Now we set $y = \cos x$, $x = \arccos y$, with $x \in [0, \pi]$, $y \in [-1, 1]$. Since for those values of x we have

$$\sin x = \sqrt{1 - \cos^2 x} = \sqrt{1 - y^2}$$

we have

$$(\arccos y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{-\sin x} = -\frac{1}{\sqrt{1-y^2}}$$

□

- Note that the functions $\arcsin x$, $\arccos x$, although defined and continuous on $[-1, 1]$, are not differentiable at the endpoints of the interval: precisely, at these points they have a vertical tangent.

The usefulness of the theorem on the derivative of the inverse function lies in the fact that it allows us to compute the derivative of the inverse function g even in situations where we do not know how to write g explicitly.

Example 9: Derivative of the inverse function

Let

$$f(x) = x + e^x$$

The function is strictly increasing on all of $\mathbb{R}$, hence invertible; let g be its inverse. We compute, for example, $g'(y_0)$ for $y_0 = f(0) = 1$. We have:

$$f'(x) = 1 + e^x, \quad f'(0) = 2 \neq 0$$

Hence:

$$g'(1) = \frac{1}{f'(0)} = \frac{1}{2}.$$

Remark 10

Given the function $f(y) = \text{SettSh } y$ (the inverse hyperbolic sine), the derivative function is $f'(y) = \frac{1}{\sqrt{y^2+1}}$

Proof. Now we set $y = \text{Sh } x$, $x = \text{SettSh } y$, with $x \in \mathbb{R}$, $y \in \mathbb{R}$. Since for those values of x we have

$$\text{Ch } x = \sqrt{\text{Sh}^2 x + 1} = \sqrt{y^2 + 1}$$

we have:

$$(\text{SettSh } y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\text{Ch } x} = \frac{1}{\sqrt{y^2 + 1}}$$

□

Remark 11

Given the function $f(y) = \text{SettCh } y$ (the inverse hyperbolic cosine), the derivative function is $f'(y) = \frac{1}{\sqrt{y^2-1}}$

Proof. Now we set $y = \text{Ch } x$, $x = \text{SettCh } y$, with $x \in [0, +\infty)$, $y \in [1, +\infty)$. Since for those values of x we have

$$\text{Sh } x = \sqrt{\text{Ch}^2 x - 1} = \sqrt{y^2 - 1}$$

we have:

$$(\text{SettCh } y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\text{Sh } x} = \frac{1}{\sqrt{y^2 - 1}}$$

□

1.4 Logarithmic derivative and elasticity

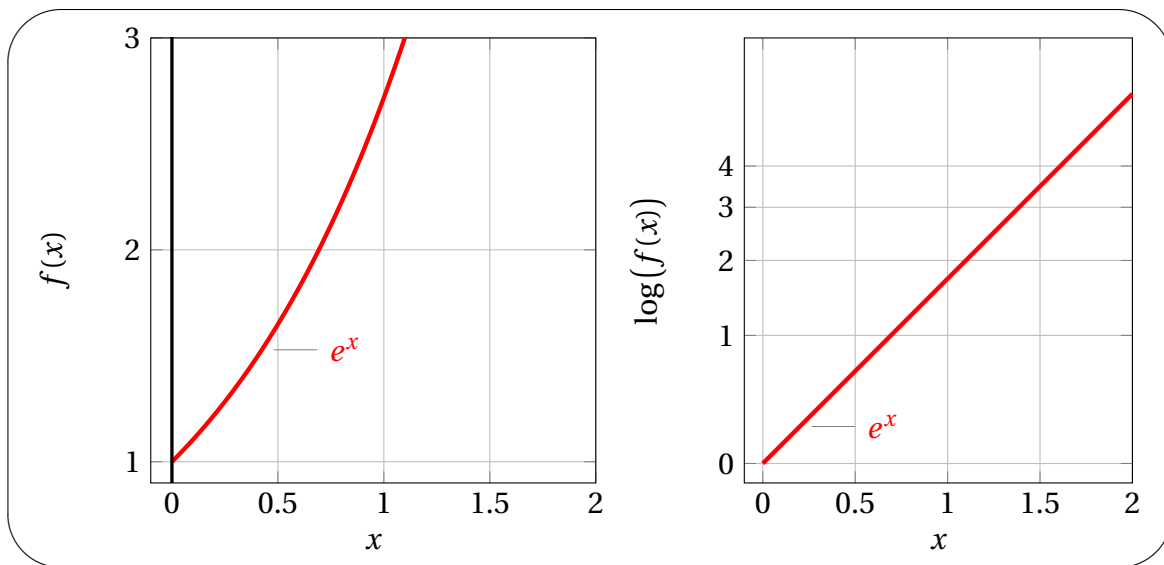
Definition 1: of logarithmic derivative

Given $f > 0$, the logarithmic derivative function of f is the derivative function of $\log f$; in formulas:

$$\frac{d}{dx} \log(f(x)) = \frac{f'(x)}{f(x)}$$

- The logarithmic derivative has the meaning of **relative growth rate** of f with respect to x .
- In some cases the relative growth rate $\frac{f'(x)}{f(x)}$ is often more meaningful than the absolute rate $f'(x)$.
- For example, a yearly capital increase of 1 billion out of a total of 10 has a very different effect from an increase of 1 billion out of 1000 billion. In the first case the relative rate of change is $1/10$, in the second $1/1000$, while the absolute increases are equal.

When we are interested in visualizing relative increases, we use graphs on a **semi-logarithmic scale**: the values of x are placed on the x -axis, while the values of $\log(f(x))$ are placed on the y -axis



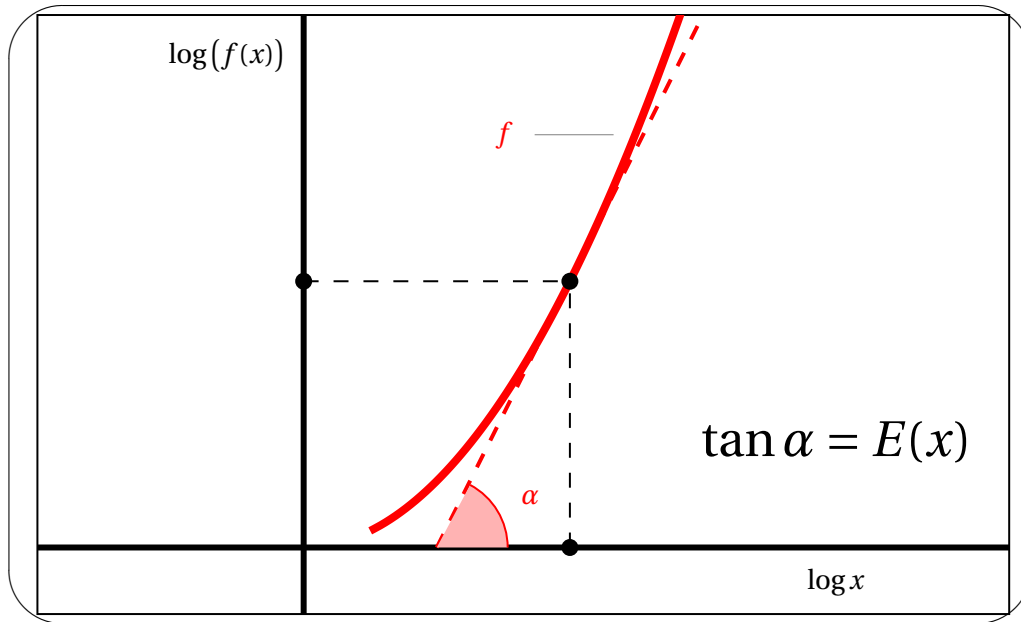
- This kind of representation is also used when $f(x)$ grows so rapidly that it would require an excessive compression of the scale (unit of measure) on the y -axis (or on the x -axis).

A **log-log graph** (graph on a logarithmic scale) is a graph in which, instead of x , the values of $\log x$ are placed on the x -axis and, instead of $f(x)$, the values of $\log f(x)$ are placed on the y -axis

Definition 2: of elasticity

The slope of the tangent line to a graph on a logarithmic scale is called the elasticity of f and is denoted by $E(x)$.

- Elasticity represents the relative rate of change of f with respect to relative changes of x , and it equals the derivative of $\log f$ with respect to $\log x$



- To find the analytic expression of $E(x)$, we observe that, by the theorem on the differentiation of composite functions, setting $u = \log x$ we have:

$$\frac{d \log f}{dx} = \frac{d \log f}{du} \frac{du}{dx} \quad \text{i.e.} \quad \frac{f'(x)}{f(x)} = E(x) \cdot \frac{1}{x}$$

from which we obtain:

$$E(x) = x \cdot \frac{f'(x)}{f(x)}$$

Example 10: elasticity

We compute the elasticity of the power function $f(x) = x^\alpha$, $x > 0$, $\alpha \in \mathbb{R}$; we have:

$$E(x) = x \cdot \frac{\alpha x^{\alpha-1}}{x^\alpha} = \alpha \quad (\text{constant elasticity})$$